

Ho Chi Minh City Housing Market Analysis

*District-Level Price Trends, Inflation Adjustment,
Geographic Drivers, and Predictive Modeling*

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Abstract

This study analyzes daily residential housing prices across nine districts in Ho Chi Minh City, Vietnam, from 2017 to 2022. Using exploratory data analysis, CPI-adjusted price indices, and district-level geographic data, we identify the structural drivers of price level and growth rate differences across the city. Our primary finding is that distance to the central business district ($r = -0.76$) and population density ($r = +0.51$) are the strongest predictors of average price, while lower-density districts exhibit systematically higher cumulative growth. District 7 exemplifies this pattern, delivering 313% real growth — the highest in the dataset — explained by its mid-range CBD proximity, low population density, and large available land area. Inflation adjustment reveals that nominal growth figures overstate real appreciation in several districts, particularly Districts 3 and 6. A supplementary forecasting experiment demonstrates that short-term price levels are predictable but 30-day price appreciation is not, with XGBoost marginally outperforming a naive zero-growth baseline.

Keywords: *housing prices, Ho Chi Minh City, exploratory data analysis, inflation adjustment, geographic determinants, time-series forecasting, XGBoost*

1. Problem Description and Goals

Residential property markets in rapidly urbanizing cities are shaped by both hyper-local and macro-level forces simultaneously. At the local level, factors such as proximity to commercial centers, land availability, and population density drive price differences between neighborhoods. At the city or national level, forces such as inflation, interest rates, and demographic growth determine the overall trajectory of prices over time. Understanding which of these forces dominates in a given market — and which districts are positioned to outperform — requires careful data analysis and honest assessment of what the available data can and cannot tell us.

This project investigates residential housing prices across nine districts in Ho Chi Minh City (HCM), Vietnam, from January 2017 to mid-2022. HCM is Vietnam's largest city and primary economic engine, with approximately 9 million residents and one of Southeast Asia's fastest-growing property markets during the study period. The nine districts analyzed — Districts 1 through 9 — span a wide range of urban typologies, from the premium central business district (District 1) to peripheral residential and light-industrial zones (District 9).

The project was motivated by three interconnected questions:

1. How much do housing prices differ across HCM districts, and what structural factors explain these differences?
2. Do nominal price growth figures accurately reflect real (inflation-adjusted) appreciation, and does inflation adjustment change the investment narrative for any districts?
3. Is it possible to reliably predict future housing prices or price growth rates using historical data and machine learning?

The primary goal of the project is to answer the first two questions rigorously through exploratory data analysis. The third question is addressed through a supplementary predictive modeling experiment, the results of which — as discussed in Sections 5 and 6 — are informative precisely because they reveal the limits of prediction from past prices alone.

Following feedback from the course instructor, the final report foregrounds the EDA findings, which are the strongest analytical contribution, and frames the predictive modeling component explicitly as exploratory, with clear disclosure of its limitations.

1.1 Datasets Used

Four datasets were incorporated:

- `HousePricingHCM.csv`: Daily housing prices for nine HCM districts, 1,363 observations per district (primary dataset).
- `consumerPriceIndex.csv`: Annual monthly-average Vietnamese CPI values from 1995 to 2024, used to deflate nominal prices to real terms with 2017 as base year.
- `populationData.csv`: World Bank national Vietnam population data, used to compute annual population growth rates.
- `district_context.csv`: District-level area (km²), population, and coordinates, used to derive population density and distance to District 1 (the CBD).

2. Team Member Roles

The project was completed as a fully collaborative effort. All three team members — Todd Klinger, Emerson Carmine, and William Griffith — contributed equally across all components of the project, including data preprocessing, exploratory data analysis, predictive modeling, figure generation, interpretation of results, and report writing. No formal division of responsibilities was maintained; decisions about methodology, framing, and conclusions were made jointly throughout the project.

3. State of the Art / Related Work

3.1 Urban Economics and Housing Price Determinants

The theoretical foundation for understanding cross-district price differences in urban housing markets draws on the classical monocentric city model developed by Alonso (1964) and Muth (1969). This framework predicts that land values and housing prices should decline monotonically with distance from the city's central business district (CBD), as higher transport costs to the center must be compensated by lower land prices for residents in peripheral locations. Our finding that distance to District 1 correlates at $r = -0.76$ with average price across HCM districts is directly consistent with this theoretical prediction.

Subsequent research has extended this framework to account for supply-side factors, including land availability, zoning restrictions, and construction activity. Glaeser and Gyourko (2003) demonstrate that differences in housing supply elasticity — driven primarily by geographic and regulatory constraints on development — are as important as demand factors in explaining cross-city price dynamics. Our finding that lower-density districts (with more available land) exhibit higher growth rates ($r = -0.52$ between density and growth) is consistent with a land availability channel operating at the district level within HCM.

For the Vietnamese context specifically, research on HCM property markets has highlighted the role of rapid urbanization, foreign direct investment, and speculative activity in driving price appreciation (Pham et al., 2020). The dramatic appreciation in peripheral districts like District 7 and 9 during our study period is consistent with the urbanization spillover dynamics documented in comparable Southeast Asian cities such as Bangkok and Jakarta.

3.2 Machine Learning for Housing Price Prediction

The application of machine learning to housing price prediction is well-established in the literature. Gradient boosting methods — including XGBoost (Chen & Guestrin, 2016) and LightGBM — consistently outperform linear models and random forests on tabular housing data with rich feature sets, particularly when the feature set includes transaction characteristics, neighborhood attributes, and macroeconomic variables.

However, the literature also documents a fundamental challenge for time-series housing price prediction: in markets where prices move smoothly, autoregressive models that rely primarily on lagged price values tend to dominate, and the apparent predictive accuracy of such models largely reflects autocorrelation rather than the identification of causal price drivers (Case & Shiller, 1989). Our results — where `lag_1` accounts for 52% of XGBoost feature importance and R^2 exceeds 0.99 — are consistent with this well-documented pattern.

For growth rate prediction specifically, the literature is considerably more pessimistic. Rapach and Zhou (2013) review the evidence on return predictability in financial markets and conclude that short-term return forecasting from past prices alone is generally unsuccessful. Housing markets are typically considered less efficient than financial markets due to illiquidity and search frictions, but the difficulty of predicting short-term price changes from lag features alone remains a consistent finding. Our result that XGBoost barely outperforms a zero-growth baseline (2.17 vs 2.18 pp MAE) is consistent with this literature.

3.3 Inflation Adjustment in Real Estate Analysis

The distinction between nominal and real price appreciation is a standard concern in real estate economics. Shiller (2015) demonstrates that long-run real U.S. housing price appreciation has been far more modest than nominal figures suggest, after accounting for general price level increases. The same methodological concern applies in the HCM context, where Vietnam experienced meaningful

inflation over the 2017–2022 period. Our finding that Districts 3 and 6 show only 19–21% real growth despite 38–40% nominal growth highlights the importance of CPI adjustment for accurate market assessment.

4. Approach

4.1 Data Preprocessing

The primary dataset was converted from wide format (one column per district) to long format (district, date, price) for analysis. Missing values were minimal and removed. The following features were constructed for the predictive modeling component:

- Lag features: lag_1, lag_2, lag_3, lag_7, lag_14, lag_30 — previous price observations at each lag distance, computed per district.
- Rolling statistics: rolling mean and standard deviation over 3, 7, 14, and 30-day windows, using a one-day shift to prevent data leakage from future observations.
- Time features: year, month, quarter, day of year.
- District identifier: one-hot encoded as a categorical variable.

For the CPI adjustment, annual monthly-average CPI values were used to construct a cumulative price level index with 2017 as the base year. This index was applied to all nominal price series to produce real (inflation-adjusted) price series. The geographic context dataset was constructed by computing population density from district area and population, and straight-line distances from each district centroid to District 1 using the haversine formula.

4.2 Exploratory Data Analysis

The EDA component involved five analytical steps:

4. Summary statistics by district: average price, standard deviation, min, max, and cumulative growth rate (both nominal and real).
5. Time-series visualization: plotting price trajectories for all nine districts over the full 2017–2022 period to identify trends, discontinuities, and comparative growth patterns.
6. Inflation adjustment: constructing real price series using the CPI index and comparing nominal versus real cumulative growth by district.
7. Geographic correlation analysis: computing Pearson correlations between district-level geographic variables (distance to CBD, population density, area) and price outcomes (average price, growth rate).
8. District clustering: applying K-means clustering ($k=3$) on a standardized feature set of price statistics and growth rate, with PCA reduction to two dimensions for visualization. An elbow plot and silhouette analysis were used to assess the robustness of the cluster choice.

4.3 Predictive Modeling

The predictive modeling component followed a consistent pipeline for all models:

- Train-test split: time-based 80/20 split, with the first 80% of observations (by date) used for training and the remaining 20% for testing. This prevents look-ahead bias.
- Pipeline: a scikit-learn Pipeline was used for each model, incorporating a ColumnTransformer that applied OneHotEncoding to the district variable and StandardScaler to all numeric features.
- Models tested: Linear Regression, Random Forest (100 trees), and XGBoost (300 estimators, learning rate 0.05, max depth 6, subsample 0.9).
- Evaluation metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 on the test set.

For the multi-horizon analysis, XGBoost was retrained with the target variable shifted by 1, 7, and 30 days respectively to assess how forecast accuracy degrades with prediction horizon.

For the growth rate prediction task, the target variable was transformed to 30-day percentage price change. XGBoost was trained on this target and compared against two naive baselines: a zero-growth baseline (always predicts 0% change) and a past-growth baseline (predicts the previous 30-day growth rate will repeat).

5. Experiments and Evaluation

5.1 District Price Levels

Table 1 summarizes the average price, cumulative nominal growth, and cumulative real growth for each district over the study period.

District	Avg Price (M VND)	Nom. Growth	Real Growth	Density (ppl/km ²)	Dist. to D1 (km)
District 1	193.2	144%	111%	18,319	0.0
District 2	67.1	91%	65%	3,750	6.1
District 3	159.7	40%	21%	37,512	2.0
District 4	93.9	30%	12%	41,220	2.2
District 5	157.0	141%	108%	37,085	5.0
District 6	88.3	38%	19%	32,640	8.1
District 7	74.8	379%	313%	10,091	5.3
District 8	77.5	112%	84%	22,194	9.8
District 9	49.9	190%	151%	4,263	14.4

Table 1: Summary statistics by district. Real growth is CPI-adjusted (2017 base). District 7 highlighted as the growth outlier.

The results confirm that price levels correlate strongly with distance to the CBD ($r = -0.76$) and with population density ($r = +0.51$). The inflation adjustment reveals that Districts 3 and 6 — which show modest nominal growth of 38–40% — experienced real growth of only 19–21%, confirming that their apparent appreciation largely reflects general inflation rather than genuine housing market dynamics.

5.2 Price-Level Prediction Results

Model	MAE (M VND)	RMSE (M VND)	R ²
Linear Regression	1.53	1.90	0.9989
XGBoost	1.58	2.09	0.9986
Random Forest	1.79	2.59	0.9979

Table 2: Price-level prediction results on the held-out test set.

All three models achieve $R^2 > 0.99$. Linear Regression performs best on both MAE and RMSE. Feature importance analysis reveals that lag_1 (yesterday’s price) accounts for 52% of XGBoost’s model weight, and all lag features combined account for 58%. This confirms that the high accuracy primarily reflects the autocorrelated nature of daily housing prices rather than the discovery of causal predictors.

5.3 Multi-Horizon Forecast Evaluation

Horizon	MAE (M VND)	RMSE (M VND)	R ²	MAE Increase vs 1-Day
1-Day	1.65	2.16	0.9985	—
7-Day	2.05	2.84	0.9975	+24%
30-Day	2.79	4.01	0.9950	+69%

Table 3: XGBoost accuracy at different forecast horizons.

Forecast accuracy degrades substantially with horizon. MAE grows 69% moving from 1-day to 30-day prediction. At the 30-day horizon, several districts show negative R² values (Districts 4, 5, 6, 7, 8), indicating that the model performs worse than simply predicting the district’s mean price for those markets.

5.4 Growth Rate Prediction

Model / Baseline	MAE (percentage points)	vs Zero-Growth Baseline
XGBoost	2.17	−0.01 pp (marginally better)
Zero-growth baseline	2.18	—
Past-growth baseline	3.31	+1.13 pp (worse)

Table 4: 30-day growth rate prediction results.

XGBoost marginally outperforms the zero-growth baseline by 0.01 percentage points — a difference with no practical significance. The growth prediction task is effectively unsolvable from lag features alone, which is itself a substantive finding about the structure of the HCM housing market.

5.5 Residual Analysis

Residual analysis of the Linear Regression model reveals three key patterns. First, the residual distribution is approximately normal and centered at −0.27M VND, indicating a small but systematic over-prediction bias. Second, District 1 has the highest absolute MAE (1.82M VND), reflecting the higher price volatility of premium districts. Third, District 9 has the worst mean percentage error (2.3%) despite having a moderate absolute MAE (1.53M), because its low average price amplifies small absolute errors into large percentage errors. This highlights the importance of reporting percentage alongside absolute errors when comparing districts with different price levels.

6. Discussion of Results

6.1 The Geographic Story

The clearest and most interpretable finding in the project is that geography — specifically distance to District 1 and population density — explains a substantial portion of cross-district price variation. Distance to CBD alone accounts for approximately 58% of variance in average price across districts ($r^2 = 0.58$), consistent with the urban economics theoretical prediction.

District 7 is the most analytically interesting case. Its combination of mid-range CBD distance (5.3 km), the lowest population density of any inner district (10,091 people/km²), and the largest district area (35.7 km²) created conditions for exceptional price appreciation. Low density signaled large available land for development; mid-range distance positioned it to absorb demand overspill from premium central districts as those became unaffordable. Growth from a low starting price across a large land area produces dramatic percentage gains even with moderate absolute price increases per square meter. This structural explanation is grounded in urban economic theory and directly supported by the data.

Primary Finding

District 7's 313% real growth over 2017–2022 reflects a specific and interpretable combination of geographic conditions, not random market variation. This is the strongest original finding of the project — a genuine insight that would not be apparent from examining price levels alone.

6.2 What Inflation Adjustment Reveals

The inflation adjustment analysis changes the narrative for several districts substantially. Districts 3 and 6 appear to be moderate growth markets on nominal figures (38–40%) but are essentially flat in real terms (19–21%), meaning an investor in those markets barely kept pace with general inflation. By contrast, District 7's exceptional nominal growth (379%) reflects genuine real appreciation (313%), confirming that its gains are not an artifact of the price level environment.

This finding matters because housing market analysis — including in academic and policy contexts — frequently relies on nominal figures. The practice of reporting nominal growth without CPI adjustment can systematically overstate the real performance of markets with moderate absolute price levels, leading to misleading comparisons between districts.

6.3 What the Predictive Models Tell Us

The predictive modeling results are informative primarily for what they reveal about the limits of prediction from past prices. The $R^2 > 0.99$ for price-level models is not evidence of a powerful forecasting system — it reflects the autocorrelated, smooth nature of daily housing prices. Any model that learns that tomorrow's price is close to today's price will achieve high accuracy on this task. The dominance of lag_1 in feature importance (52% of model weight) confirms this interpretation.

The growth prediction result — where XGBoost barely outperforms a zero-growth baseline — is the most honest and important modeling finding. It tells us that the direction and magnitude of short-term price change cannot be reliably forecast from historical prices in this market. This is consistent with the efficient market hypothesis applied to housing, and it motivates future work incorporating supply-side and macroeconomic variables.

The comparison of models is also instructive: Linear Regression outperforms Random Forest and XGBoost for price-level prediction, which is consistent with the dominant relationship being nearly linear (tomorrow's price $\approx$ yesterday's price). The more complex models offer no benefit when the underlying pattern is simple.

6.4 Price Inequality

The absolute price gap between the most and least expensive districts grew from approximately 85M VND in 2017 to 147M VND in 2022, a 74% increase. However, the coefficient of variation — which measures relative dispersion — remained flat at 0.43–0.50 throughout the period. The interpretation is that the premium tier (Districts 1, 3, 5) is pulling ahead in absolute terms simply because its prices are higher to begin with, not because the proportional structure of the market is changing. Relative inequality is stable; absolute divergence is real. Which of these is the more policy-relevant framing depends on the question being asked.

7. Lessons Learned About Machine Learning

7.1 High R^2 Is Not the Same as a Good Model

The most important lesson from this project is that impressive-looking metrics can be deeply misleading. Our price-level models achieved $R^2 > 0.99$, which would typically indicate exceptional performance. But when we examined feature importance, we found that the model was essentially learning a single relationship: yesterday's price predicts tomorrow's price. A model that learns this is not doing anything analytically interesting — it is exploiting autocorrelation, not identifying causal drivers of housing prices. This taught us to always interrogate model internals, not just headline metrics.

7.2 The Difference Between Predicting Levels and Predicting Changes

We learned that predicting price levels and predicting price changes are fundamentally different problems with very different degrees of difficulty. Level prediction is easy when the series is autocorrelated: the model can simply anchor on the recent value. Change prediction requires identifying what drives the direction and magnitude of movement, which is a much harder problem that requires causal variables rather than lag features. The practical implication is that level prediction, despite its high apparent accuracy, is often the less useful of the two for decision-making, because a decision-maker cares about where the market is going, not just where it currently is.

7.3 Feature Importance Is as Important as Prediction Accuracy

Before examining feature importance, the project appeared to have a strong predictive model. After examining feature importance, we understood that the model was doing something much simpler and less interesting. Feature importance analysis transformed our interpretation of the project entirely. This reinforced the importance of model interpretability as a component of machine learning analysis, not an optional add-on.

7.4 Baseline Comparisons Are Essential

The growth rate prediction experiment was most informative when we compared XGBoost to the zero-growth baseline. Without that comparison, a result of 2.17 pp MAE might seem acceptable. With it, the result is clearly no better than doing nothing. This taught us that the value of a predictive model is always relative to what a simple, naive alternative can achieve, and that a good evaluation always includes at least one naive baseline.

7.5 Model Choice Should Match Data Structure

The fact that Linear Regression outperformed Random Forest and XGBoost for price-level prediction was initially surprising. In retrospect it makes sense: when the dominant predictive relationship is approximately linear (tomorrow's price $\approx$ yesterday's price), a linear model will match or outperform more flexible models that can overfit to noise. Simpler models are sometimes better, and the choice of model should be guided by understanding the structure of the data rather than a prior belief that more complex models are always superior.

7.6 External Variables Matter

Throughout the project, we found that the variables we had — past prices, time features, district identifier — were insufficient for the most interesting prediction task (growth rate forecasting). The EDA consistently suggested that supply-side factors (available land, density), macroeconomic factors (interest rates, credit conditions), and infrastructure investment are likely to be the real drivers of price appreciation. This taught us that defining the right feature set is often more important than selecting

the right model, and that collecting richer data is frequently more valuable than trying more sophisticated algorithms on the same data.

8. Challenges Faced and Goals Not Achieved

8.1 Challenges Faced

Data discontinuities of uncertain origin. Several district price series contain sharp, discrete jumps — most notably District 1 around early 2019 (a spike to approximately 283M VND) and District 4 in late 2019. The source of these discontinuities is unclear — they could reflect genuine market events, policy changes, data collection methodology changes, or errors. These discontinuities complicated model training and evaluation, as they represent structural breaks that are difficult to anticipate from smooth lag features.

CPI data format. The consumer price index file contained a null byte at the end, non-standard quoting, and a structure that required careful parsing to extract the year-value pairs correctly. This was more time-consuming to resolve than expected and highlighted the importance of data quality checks early in a project.

Confounding of geographic variables. Population density and distance to CBD are moderately correlated with each other ($r = -0.51$), which means their independent effects on price cannot be separated from the bivariate correlations alone. We recognized that a proper multivariate regression controlling for both simultaneously would be needed to isolate partial effects, but implementing and interpreting this correctly within the project timeline proved challenging.

Growth rate prediction. We invested significant effort in trying to improve the 30-day growth rate prediction model — testing different feature combinations, lag structures, and hyperparameter settings. Despite this effort, we were unable to meaningfully outperform the zero-growth baseline. While this result is informative, it was frustrating to invest modeling effort without achieving a usable forecast.

Cluster count justification. The silhouette analysis suggested $k=2$ clusters, but we felt that $k=3$ better captured the analytical distinction between the premium tier and the mid-market group versus District 7 as a growth outlier. Reconciling statistical optimality ($k=2$) with analytical interpretability ($k=3$) required careful discussion and transparent disclosure.

8.2 Goals Not Achieved

District-level population dynamics. We had hoped to incorporate district-level population change data over 2017–2022 to investigate how demographic shifts within the city contributed to price appreciation in different districts. The World Bank dataset provides only national-level population figures, and district-level data from the General Statistics Office of Vietnam was not accessible within the project timeline. This remains the most significant gap in the analysis.

Supply-side variables. Construction permits, housing completions, land availability data, and transaction volume figures would substantially strengthen both the EDA and the predictive modeling components. The EDA suggests that supply-side factors are likely the primary drivers of growth rate differences between districts, but we were unable to obtain this data. Future work should prioritize acquiring these variables.

Interest rate and macroeconomic integration. We originally planned to incorporate Vietnam's benchmark interest rates as a feature in the predictive models, motivated by the expectation that credit availability would affect housing demand and prices. This data was available from the State Bank of Vietnam but integrating it correctly with the daily price series — accounting for the lag between rate changes and market responses — proved more complex than anticipated and was ultimately deferred.

Formal causal identification. The geographic analysis identifies strong correlations between distance, density, and price outcomes, but correlation is not causation. Properly establishing a causal link — for example, using instrumental variable methods or difference-in-differences designs exploiting infrastructure investments — would require research designs beyond the scope of this term project.

9. Conclusions

This project analyzed daily residential housing prices across nine Ho Chi Minh City districts from 2017 to 2022. The EDA is the primary contribution, producing three findings we consider genuinely new and interpretable.

First, District 7's 313% real growth has a structural geographic explanation: its combination of mid-range CBD proximity, low population density indicating available land, and large district area created conditions for exceptional price appreciation driven by urban expansion spillover dynamics.

Second, inflation adjustment materially changes the narrative for several districts. Districts 3 and 6 show only 19–21% real growth despite 38–40% nominal growth. Nominal figures overstate real appreciation for these markets and should not be used as the basis for investment comparisons.

Third, short-term price appreciation is not reliably predictable from past prices alone. XGBoost barely outperforms a zero-growth baseline for 30-day growth prediction, confirming that supply-side and macroeconomic variables are necessary for meaningful growth forecasting.

Overall Conclusion

Ho Chi Minh City housing prices show strong short-term persistence, making near-term price-level prediction technically accurate but analytically shallow. The more meaningful contribution of this project is the district-level EDA: geography — specifically distance to the CBD and population density — is the strongest structural predictor of both price levels and growth trajectories. District 7 exemplifies how geographic factors interact to produce exceptional market performance and provides a case study in the structural analysis that complements time-series modeling in urban housing research.

References

- Alonso, W. (1964). *Location and land use: Toward a general theory of land rent*. Harvard University Press.
- Case, K. E., & Shiller, R. J. (1989). The efficiency of the market for single-family homes. *American Economic Review*, 79(1), 125–137.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- General Statistics Office of Vietnam. (2024). *Monthly consumer price index by months and year, 1995–2024*. Hanoi: GSO.
- General Statistics Office of Vietnam. (2023). *District-level housing price data, Ho Chi Minh City, 2017–2022*. Hanoi: GSO.
- Glaeser, E. L., & Gyourko, J. (2003). The impact of building restrictions on housing affordability. *FRBNY Economic Policy Review*, 9(2), 21–39.
- Ho Chi Minh City Department of Natural Resources and Environment. (2024). *Spatial cost of living index among provinces (Ha Noi = 100), 2011–2024*.
- Muth, R. F. (1969). *Cities and housing: The spatial pattern of urban residential land use*. University of Chicago Press.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Pham, T. A., et al. (2020). Housing price dynamics in Ho Chi Minh City: Evidence from district-level data. *Journal of Asian Economics*, 70, 101–215.
- Rapach, D. E., & Zhou, G. (2013). Forecasting stock returns. In G. Elliott & A. Timmermann (Eds.), *Handbook of Economic Forecasting* (Vol. 2, pp. 328–383). Elsevier.
- Shiller, R. J. (2015). *Irrational exuberance* (3rd ed.). Princeton University Press.
- World Bank. (2026). *World Development Indicators: Population, total — Vietnam*. Washington, D.C.: World Bank. <https://data.worldbank.org>